

CODY DONAHUE

Colorado Springs, CO | cmd8592@gmail.com

SOFTWARE ENGINEERING | FULL-STACK DEVELOPMENT | SYSTEMS ENGINEERING

Software engineering professional and United States Navy veteran with hands-on experience supporting application development, systems configuration, and technical troubleshooting across Windows, Unix, and macOS environments. Built and maintained backend services, SQL databases, and CI pipelines while applying secure coding practices, structured documentation, and version control to ensure system integrity and reliability. Developed and tested Java and Python applications with expanded JUnit coverage and automated GitHub workflows to reduce defects and improve deployment consistency. Brings a strong foundation in Linux systems, networking fundamentals, cybersecurity principles, and technical problem-solving, with the ability to quickly learn new technologies and support operational IT environments. Career supported by a Bachelor of Science in Computer Science and Secret security clearance. U.S. Navy nuclear veteran with extensive leadership and quality assurance experience in mission-critical environments.

- Networking Principles
- Backend Application Support
- Database Management
- Organizational Planning
- System Administration
- Computer Diagnostics
- Process Improvement
- Technical Troubleshooting
- Technical Writing

EDUCATION

Bachelor of Science in Computer Science | Old Dominion University

Relevant Coursework:

Software Engineering | Data Structures & Algorithms | Linux Systems | Agile/SAFe | Python Programming

Database Concepts | C++ for Programmers | Object-Oriented Design | Cybersecurity Fundamentals

TECHNICAL COMPETENCIES

Languages: Java | Python Backend Development | SQL | C++ | HTML/CSS/React

OS/Platforms: Windows | Unix, macOS

CI/CD & Testing: Gradle | GitHub | JUnit | Javadoc

Tools: VSCode | Eclipse | Markdown

TECHNICAL SKILLS

Programming & Development

- Developed backend services using Python and Java, building RESTful APIs and modular applications supporting structured data workflows.

- Applied object-oriented design principles to deliver maintainable code across academic and personal software projects.
- Wrote SQL queries and managed relational databases to support application data storage and reporting.
- Built frontend components using HTML, CSS, and React to create responsive user interfaces.
- Utilized C++ and algorithmic problem-solving techniques in data structures and systems coursework.

Testing, CI/CD & Quality Assurance

- Increased code reliability by expanding JUnit test coverage across multiple Java-based projects.
- Implemented automated CI pipelines using GitHub Actions to enforce testing gates and maintain build integrity.
- Documented codebases using JavaDoc and Markdown to improve maintainability and onboarding efficiency.
- Leveraged Gradle for build automation and dependency management in Java applications.

Systems, Security & Linux

- Worked across Windows, Unix, and macOS environments to develop and test applications.
- Applied Linux systems knowledge to manage development environments and command-line workflows.
- Implemented input validation and responsible AI safeguards to protect application integrity.
- Applied cybersecurity fundamentals to ensure secure coding practices in application development.
- Maintained structured configuration control using Git for version management across projects.

Agile & Software Engineering Practices

- Applied Agile and SAFe methodologies in academic team-based software development projects.
- Translated software requirements into functional designs and technical documentation.
- Used structured sprint planning and iterative development to deliver working software increments.
- Conducted peer code reviews to improve quality, readability, and maintainability.
- Applied algorithmic analysis to optimize application performance for larger datasets.

SOFTWARE PROJECTS

EduSense – AI-Assisted Tutoring Platform

- Designed and implemented a Python backend with REST API endpoints supporting dynamic content delivery.
- Built and integrated a SQLite database to manage user and session data efficiently.
- Developed a React/CSS frontend to deliver structured user interactions and responsive design.
- Implemented custom AI model handling and input sanitization to ensure responsible educational use.
- Delivered full-stack solution independently, integrating backend, database, and frontend components.

DupDetector – Java Near-Duplicate Code Finder

- Developed tokenizer and hashing pipeline to detect near-duplicate code across repositories.
- Built a command-line interface and automated report generator to streamline analysis workflows.
- Expanded JUnit test coverage and implemented GitHub Actions CI gates to improve reliability.
- Optimized performance for large codebases, improving processing efficiency and maintainability.
- Strengthened modular architecture to enhance scalability and future feature expansion.

PROFESSIONAL EXPERIENCE

U.S. Navy – Nuclear Machinist Mate / QA Supervisor Aug 2016 – May 2024

- Led and trained teams of technicians in high-reliability environments supporting nuclear propulsion systems.
- Performed formal Quality Assurance inspections, maintenance verification, and procedural compliance.
- Authored technical documentation and enforced safety/operational standards.
- Coordinated maintenance planning and execution while supervising junior personnel.
- Developed strong troubleshooting, root-cause analysis, and process-improvement skills.